MINI PROJECT REPORT

A Mini Project Report submitted

In the Partial of Fulfillment of the Requirements for

DHV-CSD-III Lab



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CERTIFICATE

This is to certify that the mini project entitled "Impact of Feature Transformation Pipelines on Predictive Model Fairness" has been successfully completed by the following students of Second Year Data Science Engineering in CERTIFICATE partial fulfillment of the laboratory requirements for the subject Data Handling and Visualization during the academic year 2026-27

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Project Guide

Abstract

Machine learning models learn from historical data, so they can copy the biases already present in it. Preprocessing steps such as scaling, encoding, dropping the sensitive attribute or resampling are usually chosen to improve accuracy, not fairness. This project checks whether these routine choices change how fairly a model treats men and women. It matters because such models are increasingly used for decisions about people, like income, credit and hiring. The data comes from the UCI Adult Income dataset (48,842 records), downloaded directly through the ucimlrepo Python package. The task is to predict whether a person earns more than 50K, with sex as the protected attribute

Missing values were handled inside each pipeline using training data only (median for numbers, an “Unknown” category for text columns), so no test information leaked in. Five pipelines were compared using Logistic Regression: a baseline with standard scaling and one-hot encoding, MinMax scaling, ordinal encoding, sex included as an input, and undersampling across income and sex groups. All five were tested on the same 30 repeated 70/30 splits.

Results were shown with mean ± standard deviation bar charts, a forest plot of paired differences, a performance-versus-fairness scatter plot, boxplots over the 30 splits, and selection rate bars by sex.

The main finding is that no pipeline removed the unfairness. In all five, men were predicted to earn above 50K more often than women, and the disparate impact ratio fell below the 80% threshold in every one of the 30 splits. Undersampling gave the highest average F1, ordinal encoding the smallest parity gap, and the baseline the smallest equalized-odds gap. The parity gap differed by only 0.0657 across pipelines, so these choices matter less than the bias already in the data.The contribution is a more careful comparison than a single split, using paired tests with uncertainty, and extending the Valentim et al. study to a new model and to scaling.

CHAPTER 1: INTRODUCTION

1.1Background and Motivation

Machine learning models are now used to make decisions about people, such as who gets shortlisted for a job, who gets extra medical care, and who gets a loan. When these models learn from historical data, they can pick up the unfairness already present in that data. In 2018, Reuters reported that Amazon had built an experimental recruiting tool, in development since 2014, that rated candidates from one to five stars. By 2015 the company found it was not rating technical candidates in a gender-neutral way, because it had been trained on resumes submitted over ten years, most of which came from men. The tool penalized resumes containing the word “women’s” and was eventually abandoned [1].

The problem is not limited to hiring. A study published in *Science* examined a commercial algorithm used to manage the care of about 200 million people in the United States each year. At the same risk score, Black patients were considerably sicker than White patients. The authors estimated that fixing this would raise the share of Black patients receiving extra help from 17.7% to 46.5% [2]. Both cases show that a model can be accurate by usual measures and still treat groups unequally.

This is now a legal issue as well. The EU Artificial Intelligence Act lists AI used in employment and in credit scoring as high-risk, and Article 10 requires training, validation and testing data to be relevant and representative, with bias detection and mitigation in data pipelines [3]. The application date for these standalone high-risk obligations was originally 2 August 2026. The Digital Omnibus on AI, approved by Parliament on 16 June 2026 and by the Council on 29 June 2026, moved it to 2 December 2027 [4]. The delay only gives organisations more time. The obligations themselves remain

Most fairness work focuses on the model or on correcting its outputs. Everyday data preparation choices, such as how to scale numbers, how to encode categories, whether to drop the sensitive attribute and whether to rebalance the data, get far less attention, even though every project makes them. Valentim et al. studied this on the Adult Income and German Credit datasets. They found that removing the sensitive attribute is not enough to remove unfairness, and that undersampling based on the true labels is sometimes more harmful than no undersampling [5]. That study used tree-based models and did not look at feature scaling.

This project builds on that gap. Using the Adult Income dataset [6], it compares five preprocessing pipelines with a Logistic Regression classifier across 30 repeated train/test splits, so the differences are measured with uncertainty and not judged from one lucky split. The aim is to show practitioners which routine preprocessing choices actually change fairness, and which do not.

1.2 Problem Statement

Problem Statement No. 7: Algorithmic fairness is a major concern in machine learning, but most research looks at model design or at fixing biased predictions afterwards, and it largely ignores the preprocessing decisions made while transforming the data. These decisions, such as how features are scaled, binned, encoded or transformed, can introduce or increase bias without anyone noticing. This project takes a dataset with sensitive attributes, the UCI Adult Income dataset, and builds several parallel preprocessing pipelines that differ in these transformation choices, then trains the same classifiers on each one. Every pipeline is compared on performance (accuracy, F1-score) and on fairness (demographic parity

difference, equalized odds and disparate impact ratio across protected groups). The goal is a framework that links each preprocessing decision to its fairness outcome with empirical evidence, so practitioners know how to build transformation pipelines that do not discriminate by accident.

1.3 Objectives

1. Design five preprocessing pipelines for one Logistic Regression classifier.
2. Implement median and "Unknown" imputation with training-only fitting.
3. Compare standard and Min-Max scaling and compare one-hot with ordinal encoding.
4. Measure the effect of including sex as a model input while retaining it for evaluation.
5. Assess training-only undersampling of the four income-by-sex cells.
6. Evaluate predictive and fairness metrics across 30 seeded 70/30 splits.
7. Summarize paired differences from the baseline with bootstrap intervals.

1.4 Scope and Limitations

1. Data coverage and constraints

The study uses one data source, the UCI Adult Income dataset, which was extracted from the 1994 US census database. It contains 48,842 records, and the geographic scope is the United States, although the native-country column lists many origins. The data is a single snapshot with no time variable, so no trends over time can be studied, and the findings reflect census conditions of that period. Only one protected attribute, sex, is analysed, and it is treated as binary because that is how the dataset records it.

There were few practical constraints. The data is public and was downloaded in code through the ucimlrepo package, so no API keys, rate limits or access restrictions applied, but an internet connection is needed to fetch it. All experiments ran on a free Google Colab CPU, and the 150 model fits (5 pipelines on 30 splits) took about **5 min**. This limited the work to a simple classifier, Logistic Regression, and to repeated hold-out splits instead of heavier methods such as nested cross-validation or large model searches. The 30 splits overlap in their data, so the reported intervals are indicative rather than exact.

2. Techniques applied and not applied

Area	Applied in this project	Not applied
Missing values	Median for numeric columns, "Unknown" for categorical columns, fitted on training data only	Dropping rows with missing values, mode imputation, KNN imputation, iterative (MICE) imputation
Numerical scaling	Standard (z-score), MinMax	Robust scaling (dropped because capital-gain and capital-loss have zero IQR)
Categorical encoding	One-hot, ordinal	Target encoding
Data balancing	Multivariate undersampling on income × sex	Oversampling, reweighing
Classifier	Logistic Regression	Tree-based or other models
Fairness metrics	SPD, DIR, EOD, TPR/FPR gaps, NPI	Post-processing or in-training fairness methods

TABLE I. Preprocessing and modelling techniques in scope

1.5 Organization of Report

Chapter 1: introduces the research problem, objectives, research questions, and scope of the study.
Chapter 2: reviews published research on data preprocessing, algorithmic fairness, fairness metrics, and the trade-off between predictive performance and fairness.
Chapter 3: describes the dataset, experimental design, preprocessing pipelines, evaluation metrics, and repeated-split methodology.
Chapter 4: presents the implementation and experimental results, including the quantitative comparisons and visualizations.
Chapter 5: discusses the findings in relation to the research questions and existing literature.
Chapter 6: concludes the study by summarizing the main findings, contributions, limitations, and possible directions for future work.

CHAPTER 2 : LITERATURE REVIEW

2.1Existing Work

1. Foundations, definitions and metrics

Mehrabi et al. [7]. Mehrabi, Morstatter, Saxena, Lerman and Galstyan published this survey in *ACM Computing Surveys* in 2021. Instead of running experiments, they reviewed real-world applications where machine learning showed bias, listed the sources of bias in data and algorithms, and built a taxonomy of fairness definitions. Their main result is a structured map of how bias arises and how fairness is defined in different settings. As a survey, it does not test any preprocessing step, and the authors note that many directions for mitigation remain open. It gives this project its vocabulary and the reason for reporting several metrics at once.

Hardt, Price and Srebro [8]. In 2016 these authors proposed equalized odds and equal opportunity as criteria for non-discrimination in supervised learning. Using only the predictor, the true label and the protected attribute, they showed how to adjust any learned predictor to meet the criterion and tested the idea on FICO credit scores. They found that such a criterion can be satisfied by post-processing and that it rewards more accurate predictors. The authors also point out the limits of "oblivious" measures like this one, which depend only on joint statistics and not on what the features mean. Their definition is the basis for the TPR, FPR and EOD gaps used here.

Feldman et al. [9]. Feldman, Friedler, Moeller, Scheidegger and Venkatasubramanian (KDD 2015) adopted the US EEOC "80% rule" as a measure of disparate impact and linked it to classification error. They proposed testing for disparate impact by checking how well the protected attribute can be predicted from the other attributes, and described a way to repair data. With three classifiers, data certified by their method showed no disparate impact. The authors note that disparate impact is not illegal in itself, because business necessity can justify it. The 80% rule is the threshold applied to the DIR in this project.

Kamishima et al. [10]. Kamishima, Akaho, Asoh and Sakuma presented this paper at ECML PKDD 2012. They discussed three causes of unfairness and proposed a prejudice remover regularizer, a penalty term for probabilistic discriminative models such as logistic regression. They also defined the prejudice index and its normalised form (NPI), which measures the dependence between predictions and the sensitive attribute. Their approach works only with that class of models, and it is an in-processing method, not a data preparation step. NPI is one of the metrics reported in this project.

2. Preprocessing as bias mitigation

Kamiran and Calders [11]. In *Knowledge and Information Systems* (2012), Kamiran and Calders surveyed and extended three families of preprocessing methods: suppressing the sensitive attribute, massaging (changing class labels) and reweighing or resampling. They implemented them in a modified Weka and ran experiments on real-life data. Massaging and preferential sampling gave a clear drop in discrimination. By their own statement, they focus on one binary sensitive attribute and two-class problems, which is also the setting of this project.

Hort et al. [12]. Hort, Zhang, Sarro and Harman introduced Fairea at ESEC/FSE 2021, a way of benchmarking bias mitigation methods against a model-behaviour baseline. They tested 12 widely studied mitigation methods. Their results were poor in 49% of cases: 15% had a worse fairness-accuracy trade-off than the baseline, and 34% lost accuracy while bias increased. This warns that a method intended to help fairness can make things worse. It studies named mitigation methods, not everyday choices such as scaling or encoding.

3. Effect of data preparation choices on fairness

Valentim, Lourenço and Antunes [5]. This 2019 study, from the University of Coimbra, examined how data preparation steps change the fairness of software systems. They used the Adult Income and German Credit datasets with Decision Trees and Random Forests and repeated the experiments over many runs. They found that removing the sensitive attribute helps but does not remove unfairness, and that undersampling based on the label can sometimes be more harmful than no undersampling. Their study covers tree models and does not test feature scaling, and this project extends it on both points.

Biswas and Rajan [13]. At ESEC/FSE 2021, Biswas and Rajan introduced a causal way of measuring the fairness effect of each preprocessing stage in an ML pipeline, including common transformers such as StandardScaler. They found that some data transformers make the model less fair, identified patterns across categories of transformers, and showed how the fairness of single stages combines into the fairness of the full pipeline. This directly supports the idea that scaling and encoding choices can change fairness. Their work is on transformer stages in general, and this project tests a fixed set of five concrete pipelines.

Friedler et al. [14]. Friedler, Scheidegger, Venkatasubramanian, Choudhary, Hamilton and Roth (FAT* 2019) built an open benchmark comparing fairness-enhancing algorithms across several fairness measures and datasets. They found that many fairness measures correlate strongly, and that the algorithms are sensitive to changes in training-test splits and to different forms of preprocessing. Their conclusion is that fairness interventions may be more brittle than expected. This is the reason this project uses 30 repeated splits and not one.

4. Dataset validity and real-world harm

Ding et al. [15]. Ding, Hardt, Miller and Schmidt (NeurIPS 2021) examined the UCI Adult dataset, which comes from a 1994 US Census survey and is widely used in fairness research. They rebuilt a superset of it from census sources, showed idiosyncrasies that limit its external validity, and released new census-based datasets covering several years and all US states. This is a direct limitation of the Adult data used here, which is why this project reports findings for this dataset only.

Obermeyer et al. [2]. Published in *Science* in 2019, this study analysed a commercial algorithm used to manage the care of about 200 million people in the US. At the same risk score, Black patients were considerably sicker than White patients, and the authors estimated that correcting this would raise the share of Black patients receiving extra help from 17.7% to 46.5%. It shows the real-world cost of bias in a deployed system. It concerns a health-care algorithm, so it is used here as motivation and not as a methodological comparison.

2.2 Research Gap Identified

The reference paper studies preparation effects with tree learners and a different validation design. This project asks a narrower question: how do scaling, encoding, sex-as-input, and joint label-by-sex undersampling compare for Logistic Regression on Adult across repeated splits? It adds paired differences and uncertainty summaries to that focused design. It is not a new fairness method. The defensible contribution is a transparent, reproducible, dataset-specific comparison with declared metric conventions and confounds.

CHAPTER 3 : DATA ACQUISITION

3.1 Raw Data Description

One source is specified: the UCI Adult dataset via ucimlrepo. The repository describes 48,842 instances and 14 predictors from a 1994 U.S. Census extraction. The six numeric predictors named are age, fnlwgt, education-num, capital-gain, capital-loss, and hours-per-week. The categorical predictors are workclass, education, marital-status, occupation, relationship, race, sex, and native-country. The target income is mapped to positive when above $50K.

Attribute	dtype	Missing %	Unique	mean	median	std	min	max
age	integer	0.0	74	38.64	37.0	13.71	17.0	90.0
workclass	categorical	5.73	8	-	-	-	-	-
fnlwgt	integer	0.0	28523	189664.13	178144.5	105604.03	12285.0	1490400.0
education	categorical	0.0	16	-	-	-	-	-
education-num	integer	0.0	16	10.08	10.0	2.57	1.0	16.0
marital-status	categorical	0.0	7	-	-	-	-	-
occupation	categorical	5.75	14	-	-	-	-	-
relationship	categorical	0.0	6	-	-	-	-	-
race	categorical	0.0	5	-	-	-	-	-
sex	categorical	0.0	2	-	-	-	-	-
capital-gain	integer	0.0	123	1079.07	0.0	7452.02	0.0	99999.0
capital-loss	integer	0.0	99	87.5	0.0	403.0	0.0	4356.0
hours-per-week	integer	0.0	96	40.42	40.0	12.39	1.0	99.0
native-country	categorical	1.75	41	-	-	-	-	-
income	categorical	0.0	2	-	-	-	-	-

TABLE II. Summary of the raw Adult Income data (n = 48,842)

CHAPTER 4:DATA PREPROCESSING AND TRANSFORMATION

4.1Data Cleaning and Munging

- **Missing-value coding:** "?" entries were converted to true missing values so the imputers could handle them.
- **Target construction:** the income column was converted to a binary label (above 50K = 1), with about 24% positive cases.
- **Protected attribute:** sex was separated and kept for evaluation, and it was only given to the model in pipeline F4.
- **Duplicates:** the number found is 52. No rows were removed, because the dataset has no ID column and identical rows can belong to different people.
- **Row count:** all 48,842 records were kept.
- **Not applicable:** date parsing (there are no date columns), encoding or character problems, and case inconsistencies such as "Mumbai/mumbai". The categorical values are already in a standard form .

4.2Handling Missing Data

1.Missingness pattern

Only workclass, occupation and native-country have missing values (5.73%, 5.75% and 1.75% of 48,842 records). Workclass and occupation are missing together (indicator correlation 0.998), while native-country is unrelated to both. A logistic regression predicting each missingness indicator from the numeric features and sex gave a 5-fold AUC of 0.714 (workclass), 0.715 (occupation) and 0.577 (native-country). Chi-square tests showed that missing workclass and occupation are strongly linked to income ($p = 5.6 \times 10^{-76}$ and 1.3×10^{-76}) and to sex ($p = 2.6 \times 10^{-45}$ and 4.4×10^{-45}). People with these fields missing work fewer hours (31.8 against 40.9 per week) and earn above 50K less often (9.5% against 24.8%). Native-country has no link to income ($p = 0.244$) and only a weak link to sex ($p = 0.009$). Missingness is therefore unlikely to be MCAR for workclass and occupation and is most plausibly MAR. MNAR cannot be ruled out, since these tests use only observed data.

2. Visualization

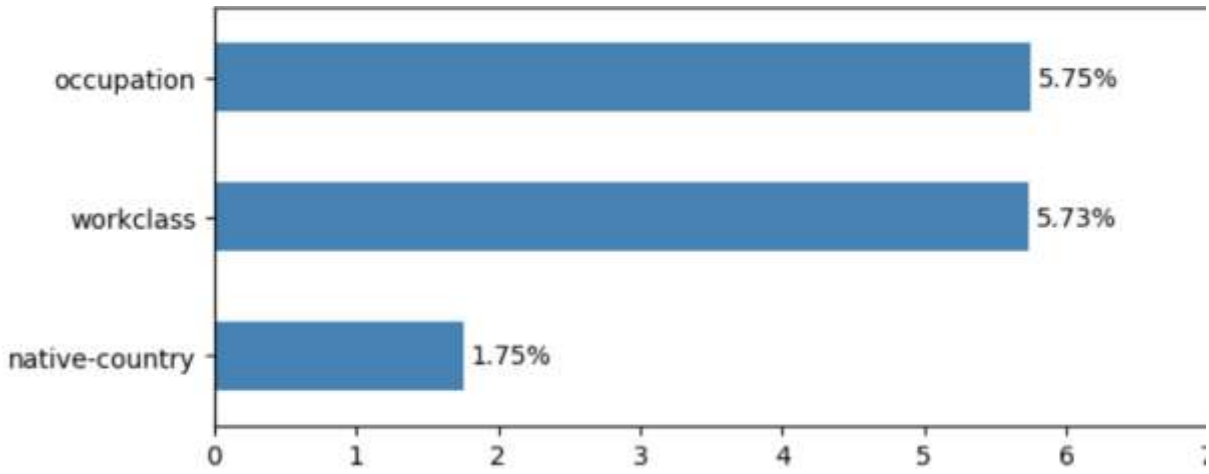


Fig. 1. Percentage of missing values per column in the raw Adult data.

4. Comparison of two methods

Column	TVD: Unknown	TVD: Mode	TVD: Random forest	Recovery: Mode	Recovery: Random forest
workclass	0.1016	0.027	0.0261	0.7342	0.7356
occupation	0.1016	0.088	0.032	0.1344	0.3136
native-country	0.1017	0.0087	0.0068	0.9143	0.922

Table. III. COMPARISON OF IMPUTATION METHODS BASED ONRECOVERY ACCURACY, TOTAL VARIATION DISTANCE, AND F1-SCORE

4.3Outlier and Anomaly Detection

Outliers were identified using Z-score (\(|z|>3\)) and IQR (1.5 × IQR) methods across six numeric columns. Grubbs' test was excluded due to its normality assumption and one-outlier-at-a-time detection. IQR was unsuitable for capital-gain and capital-loss because their zero IQR flags all non-zero values. Robust scaling was therefore excluded.All flagged values were retained without removal or capping, as they may be genuine and removing them could affect sex distribution and fair pipeline comparisons. Since no values were modified, before-and-after comparison was unnecessary.

4.4Data Transformation Techniques

Standard scaling was used in F1, F3, F4 and F5, while F2 used min-max scaling to bring features to a similar range for logistic regression. Robust scaling was excluded because capital-gain and capital-loss have zero IQR. No skew transformations, such as log, Box-Cox or Yeo-Johnson, were applied, as they were outside the study's scope. Similarly, no binning or reshaping operations were performed, and continuous variables were retained as given. One-hot encoding was used in F1, F2, F4 and F5, while F3 used ordinal encoding followed by standardization. Ordinal encoding was included for comparison, although it imposes an arbitrary order on categories. Additionally, F5 applied multivariate undersampling to balance income–sex groups in the training data, while F4 changed whether sex was included as a model input to examine its effect on fairness.

ID	Name	Numeric scaler	Categorical encoding	`sex` as model input?
F1	Baseline	Standard	One-hot	No
F2	MinMax	MinMax	One-hot	No
F3	Ordinal	Standard	Ordinal (+ StandardScaler on codes)	No
F4	Sex-as-input	Standard	One-hot	**Yes**
F5	Multivariate undersampling	Standard	One-hot	No

Table. IV. Transformations applied in each pipeline.

4.5 Feature Engineering

No new predictive features were created. This was deliberate: adding derived features would change several things at once and make it impossible to credit differences to scaling, encoding or sampling. Three helper variables were derived for experiment control only and were not given to the model: the binary income target, the sex array used for evaluation, and the income × sex key used for stratified splitting and for undersampling in F5.

CHAPTER 5:DATA VISUALIZATION AND ANALYSIS

5.1Visualization Design Rationale

- **Strategy:** the study asks comparison questions (which pipeline is better, by how much, and is the difference real), so every chart is chosen to answer one of them.
- **Grouped bars (mean ± std):** best for comparing a few pipelines on several metrics side by side. Position along a common scale is the most accurate way for people to compare values [17], [19].
- **Forest plot of paired differences:** shows each pipeline's difference from the baseline with its 95% interval, so the reader sees at once whether the interval crosses zero.
- **Scatter plot (F1 against parity gap):** the natural chart for a trade-off between two measures.
- **Box plots over 30 splits:** show spread and overlap, which a bar of means would hide.
- **Selection rate by sex:** a simple bar chart makes the gap between groups visible without decoration.

5.2 Exploratory Visualizations

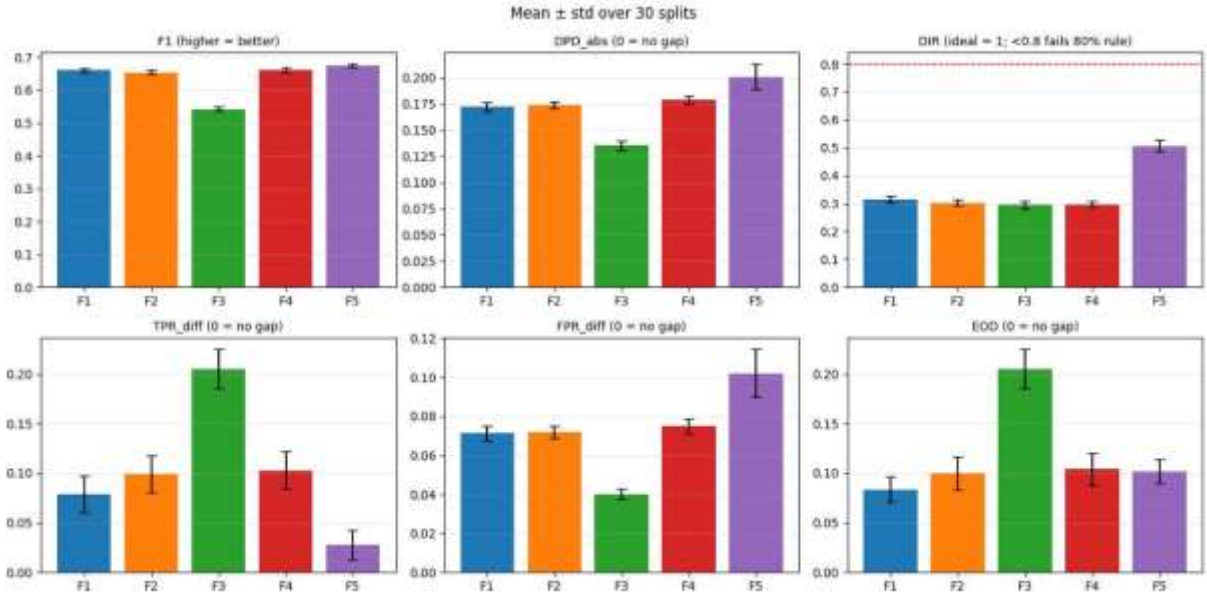


Fig. 2. Mean ± standard deviation of performance and fairness metrics for pipelines F1–F5 over 30 splits.

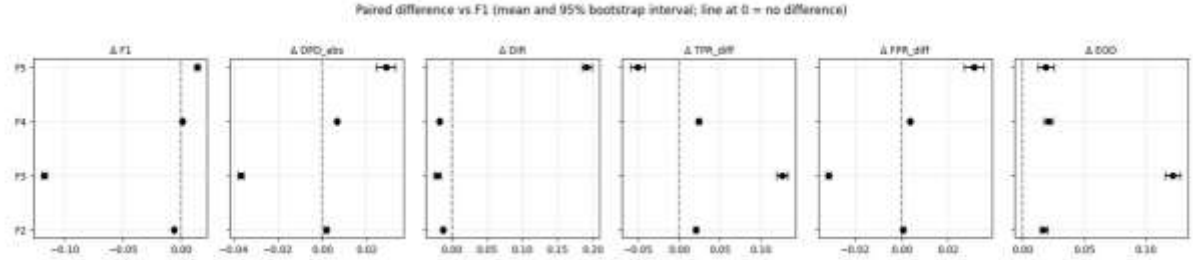


Fig. 3. Paired differences between each pipeline and the baseline F1, with 95% bootstrap confidence intervals.

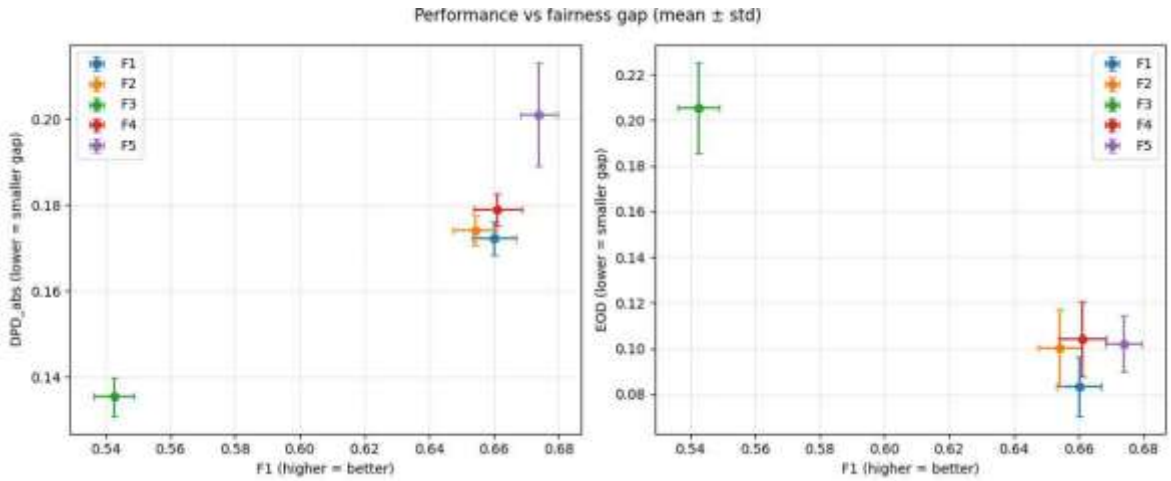


Fig. 4. Trade-off between predictive performance (F1-score) and fairness gap (DPD) across pipelines.

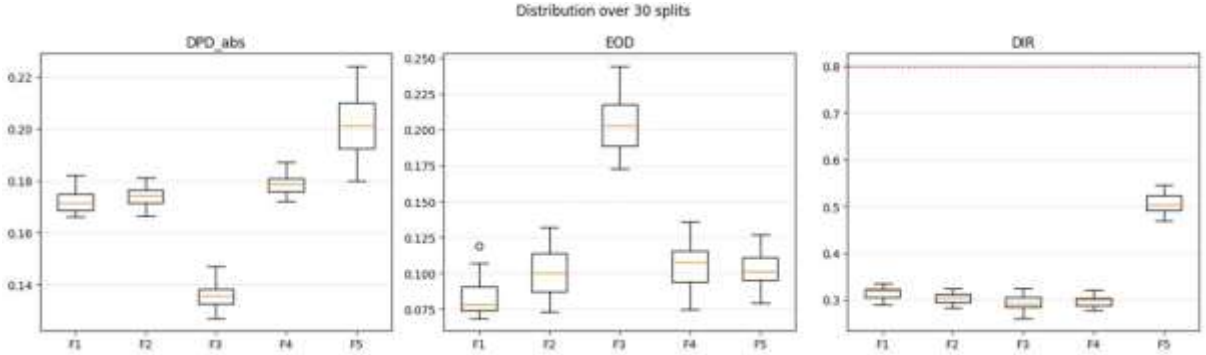


Fig. 5. DPD_abs, EOD and DIR,distribution over 30 split.

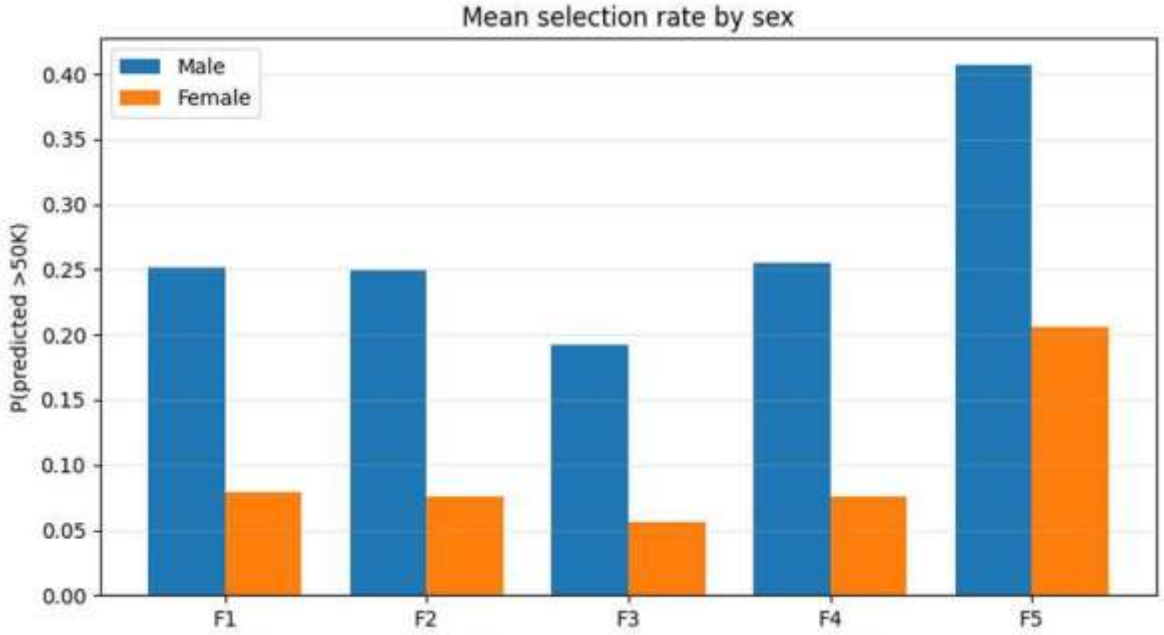


Fig. 6. Rate of predicted income above 50K for men and women under each pipeline.

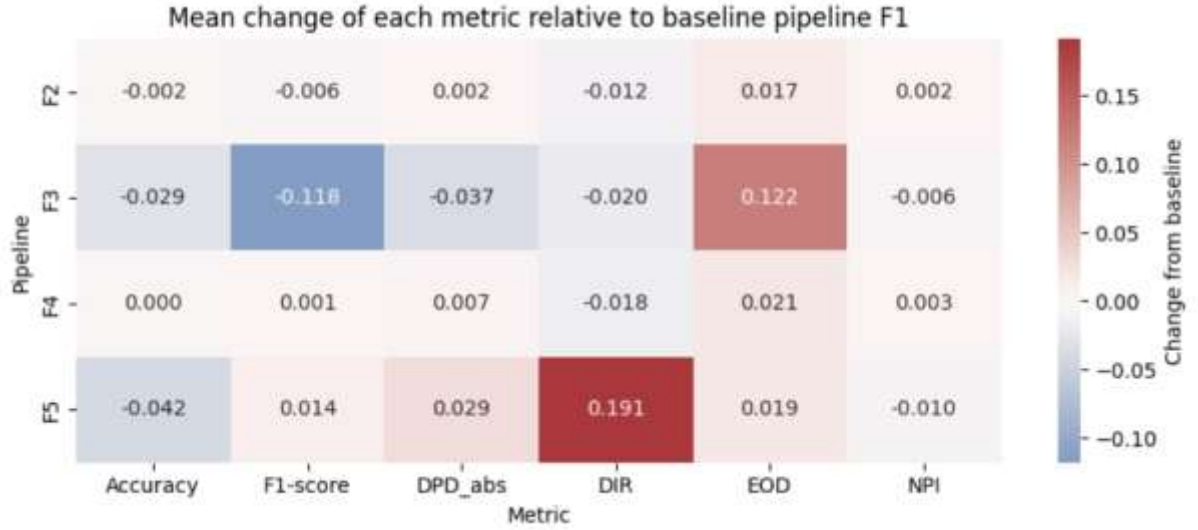


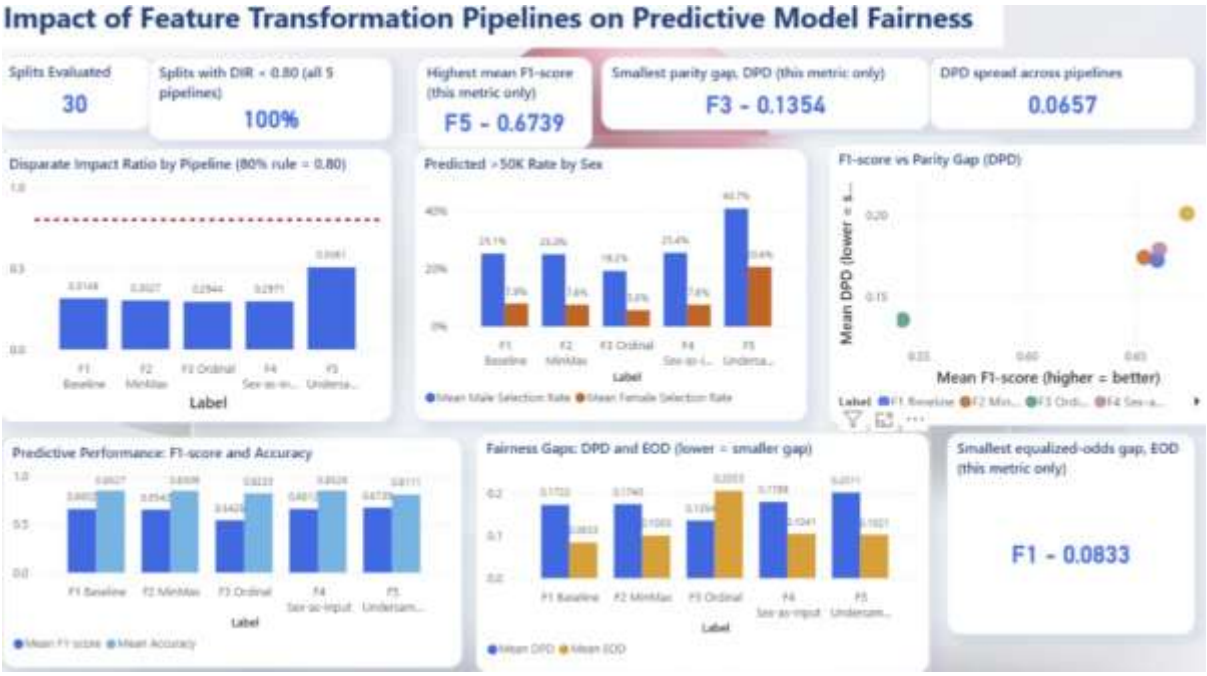
Fig.7. Mean change in each performance and fairness metric for pipelines F2–F5 relative to the baseline F1, averaged over 30 splits.

5.3Dashboard Development

1.Overall dashboard layout and design philosophy

The dashboard is a single page that follows the questions of the study. Summary cards give the average F1-score, parity gap and DIR for the selected pipelines, and the charts below compare performance, fairness gaps, the 80% rule, selection rates by sex and the performance–fairness trade-off. Each pipeline keeps the same colour throughout, all bars start at zero, and a slicer lets the reader filter by pipeline.

2.Dashboard





ID	Mean Difference	CI Low	CI High	CI Excludes Zero
F1				No
F2	0.0079	-0.0137	0.0229	No
F3	-0.2962	-0.1627	0.1335	No
F4	0.0497	-0.0191	0.0266	No
F5	0.5472	-0.1654	0.2217	No
Total	**0.3087**	**-0.1654**	**0.2217**	**No**

ID	Accuracy	DIR	DPD_abs	EOD	F1	FPR_diff	NPI	TPR_diff	Total
F2	-0.0018	-0.0120	0.0018	0.0169	-0.0060	0.0006	0.0017	0.0206	**0.0218**
F3	-0.0294	-0.0203	-0.0368	0.1220	-0.1177	-0.0314	-0.0063	0.1266	**0.0068**
F4	0.0001	-0.0177	0.0067	0.0208	0.0010	0.0036	0.0034	0.0243	**0.0421**
F5	-0.0416	0.1913	0.0288	0.0188	0.0138	0.0309	-0.0095	-0.0513	**0.1811**
Total	**-0.0726**	**0.1413**	**0.0004**	**0.1784**	**-0.1089**	**0.0038**	**-0.0107**	**0.1202**	**0.2519**

Label	Mean Accuracy	Mean Precision	Mean Recall	Mean F1-score	Mean DPD	Mean DIR	Mean TPR Gap	Mean FPR Gap	Mean EOD	Mean NPI
F1 Baseline	0.8527	0.7368	0.5980	0.6602	0.1722	0.3148	0.0787	0.0713	0.0833	0.0425
F2 MinMax	0.8509	0.7352	0.5894	0.6542	0.1740	0.3027	0.0993	0.0719	0.1003	0.0441
F3 Ordinal	0.8233	0.7130	0.4379	0.5425	0.1354	0.2944	0.2053	0.0399	0.2053	0.0362
F4 Sex-as-input	0.8528	0.7360	0.6002	0.6612	0.1789	0.2971	0.1030	0.0749	0.1041	0.0459
F5 Undersampling	0.8111	0.5741	0.8160	0.6739	0.2011	0.5061	0.0274	0.1021	0.1021	0.0329
Total	**0.8382**	**0.6990**	**0.6083**	**0.6384**	**0.1723**	**0.3430**	**0.1027**	**0.0720**	**0.1190**	**0.0403**

5.4 Key Findings from Visual Analysis

1. Bias persists across all pipelines: Men were more frequently predicted to earn above $50K than women in every pipeline.
2. The 80% rule fails: The DIR stayed below 0.8 in all 30 splits for each of the five pipelines, indicating persistent disparity.
3. Fairness changes are modest: The difference in mean DPD_abs between the best and worst pipelines was only 0.0657.
4. Performance varies more than fairness: F1 scores differed by 0.1314 across pipelines, roughly twice the spread in parity gaps.
5. Undersampling increases F1: F5 achieved the highest mean F1, possibly because balancing training labels changes the effective decision threshold, not necessarily because the model is better.
6. Ordinal encoding improves parity: F3 recorded the smallest mean DPD_abs.
7. F1 had the smallest mean equalized-odds gap

CHAPTER 6 : RESULTS AND DISCUSSION

6.1 Summary of Results

ID	Pipeline	Accuracy	Precision	Recall	F1	DIR<0.8 in splits
F1	F1 Baseline	0.8527 ± 0.0029	0.7368 ± 0.0092	0.5980 ± 0.0082	0.6602 ± 0.0068	30/30
F2	F2 MinMax	0.8509 ± 0.0029	0.7352 ± 0.0085	0.5894 ± 0.0077	0.6542 ± 0.0068	30/30
F3	F3 Ordinal	0.8233 ± 0.0021	0.7130 ± 0.0078	0.4379 ± 0.0071	0.5425 ± 0.0063	30/30
F4	F4 Sex-as-input	0.8528 ± 0.0032	0.7360 ± 0.0095	0.6002 ± 0.0087	0.6612 ± 0.0073	30/30
F5	F5 Multivariate undersampling	0.8111 ± 0.0037	0.5741 ± 0.0060	0.8160 ± 0.0110	0.6739 ± 0.0057	30/30

ID	Pipeline	SelRate_Male	SelRate_Female	SPD_signed	DPD_abs	DIR	TPR_diff	FPR_diff	EOD	NPI
F1	F1 Baseline	0.2513 ± 0.0039	0.0791 ± 0.0032	0.1722 ± 0.0039	0.1722 ± 0.0039	0.3148 ± 0.0115	0.0787 ± 0.0185	0.0713 ± 0.0037	0.0833 ± 0.0129	0.0425 ± 0.0019
F2	F2 MinMax	0.2495 ± 0.0030	0.0755 ± 0.0031	0.1740 ± 0.0035	0.1740 ± 0.0035	0.3027 ± 0.0117	0.0993 ± 0.0187	0.0719 ± 0.0030	0.1003 ± 0.0169	0.0441 ± 0.0019
F3	F3 Ordinal	0.1918 ± 0.0037	0.0565 ± 0.0027	0.1354 ± 0.0045	0.1354 ± 0.0045	0.2944 ± 0.0149	0.2053 ± 0.0199	0.0399 ± 0.0027	0.2053 ± 0.0199	0.0362 ± 0.0023
F4	F4 Sex-as-input	0.2545 ± 0.0039	0.0756 ± 0.0032	0.1789 ± 0.0038	0.1789 ± 0.0038	0.2971 ± 0.0111	0.1030 ± 0.0188	0.0749 ± 0.0038	0.1041 ± 0.0166	0.0459 ± 0.0019
F5	F5 Multivariate undersampling	0.4068 ± 0.0091	0.2057 ± 0.0055	0.2011 ± 0.0121	0.2011 ± 0.0121	0.5061 ± 0.0204	0.0274 ± 0.0148	0.1021 ± 0.0124	0.1021 ± 0.0124	0.0329 ± 0.0038

Table. V. Performance and fairness of the five pipelines (mean ± std over 30 splits).

ID	n_train	train_pos_rate	train_female_share	train_label_SPD
F1	34189.0	0.2393	0.3315	0.1946
F2	34189.0	0.2393	0.3315	0.1946
F3	34189.0	0.2393	0.3315	0.1946
F4	34189.0	0.2393	0.3315	0.1946
F5	4952.0	0.5000	0.5000	0.0000

Table. VI. TRAINING-DATA STATISTICS UNDER EACH PIPELINE (MEAN OVER 30 SPLITS)

6.2 Comparative Analysis

- **Baseline:** F1 is the baseline for all comparisons, and each other pipeline differs from it in one main choice.
- **Method:** for each pipeline, the per-split difference from F1 was computed, and a 95% bootstrap interval (2,000 resamples) was built [20]. A difference counts as real if the interval excludes zero.

- **Why paired:** all pipelines use the same 30 splits, which removes split-to-split noise.
- **Comparison with the literature:**
  - Like Valentim et al. [5], removing sex does not remove unfairness.
  - Like Friedler et al. [14], results vary with the split, which is why 30 splits were used.
  - Biswas and Rajan [13] suggest scalers can matter.

metric ID	F1	DPD_abs	DIR	TPR_diff	FPR_diff	EOD
F2	-0.0060 [-0.0068, -0.0053]	+0.0018 [+0.0010, +0.0025]	-0.0120 [-0.0137, -0.0103]	+0.0206 [+0.0183, +0.0229]	+0.0006 [+0.0000, +0.0013]	+0.0169 [+0.0140, +0.0201]
F3	-0.1177 [-0.1198, -0.1155]	-0.0368 [-0.0380, -0.0356]	-0.0203 [-0.0251, -0.0154]	+0.1266 [+0.1199, +0.1335]	-0.0314 [-0.0322, -0.0304]	+0.1220 [+0.1155, +0.1279]
F4	+0.0010 [+0.0005, +0.0015]	+0.0067 [+0.0062, +0.0071]	-0.0177 [-0.0191, -0.0164]	+0.0243 [+0.0219, +0.0266]	+0.0036 [+0.0033, +0.0040]	+0.0208 [+0.0174, +0.0242]
F5	+0.0138 [+0.0118, +0.0158]	+0.0288 [+0.0244, +0.0330]	+0.1913 [+0.1844, +0.1986]	-0.0513 [-0.0599, -0.0428]	+0.0309 [+0.0267, +0.0353]	+0.0188 [+0.0121, +0.0252]

Table. VII. PAIRED DIFFERENCE FROM F1.

6.3 Discussion and Interpretation

The results suggest that data characteristics contribute more to unfairness than preprocessing choices. Even when sex is removed, correlated features such as relationship, marital status, and occupation may indirectly preserve gender-related bias [10], [5]. Preprocessing produces only modest improvements because the Adult dataset already reflects income differences between men and women, which the linear model can reproduce.
F5 achieved the highest F1 score, but this does not necessarily mean better fairness or a better model. Undersampling changes the training label balance, potentially increasing positive predictions and recall. Similarly, F3 showed the smallest parity gap, but this may result from information loss due to ordinal encoding rather than fairer treatment.
Overall, selecting a scaler or encoder based only on accuracy is unlikely to eliminate bias. Fairness must be measured and addressed separately using appropriate metrics and mitigation methods [8], [11]. The variation across data splits supports the need for repeated evaluation [14], while the possibility of unintended effects highlights the limitations of fairness interventions [12]. Finally, since the Adult dataset dates back to 1994, its findings may not fully generalise to modern populations and employment conditions [15].

6.4 Contribution

This project does not propose a new fairness algorithm. Its contribution is a controlled, paired comparison of five common preprocessing pipelines on the same 30 repeated splits, with bootstrap intervals for each difference. It extends earlier work on data preparation and fairness [5], [13] to a new classifier (Logistic Regression) and to feature scaling. It shows that on the Adult data these choices shifted the parity gap by at most 0.0657, while every pipeline failed the 80% rule in all 30 splits.

CHAPTER 7 : CONCLUSION AND FUTURE WORK

7.1Conclusion

Machine learning models trained on historical data can treat groups of people unequally, and the everyday preprocessing steps applied before training are rarely checked for their effect on fairness. This project set out to measure how much those routine choices matter. Using the UCI Adult Income dataset, where the task is to predict whether income exceeds 50K and sex is the protected attribute, we compared five preprocessing pipelines with the same Logistic Regression classifier: standard scaling with one-hot encoding as the baseline, min-max scaling, ordinal encoding, sex included as an input, and multivariate undersampling over income and sex groups.

To keep the comparison reliable, every pipeline was trained and tested on the same 30 repeated stratified 70/30 splits, and all fitting steps, including imputation, were done on training data only. Performance was measured with accuracy, precision, recall and F1-score, and fairness with statistical parity difference, disparate impact ratio, equalized-odds gaps and the normalised prejudice index. Differences from the baseline were estimated per split and summarised with 95% bootstrap intervals.

The main finding is that no preprocessing choice made the model fair. In all five pipelines men were predicted to earn above 50K more often than women, and the disparate impact ratio fell below the 80% threshold in every one of the 30 splits. Undersampling gave the highest average F1-score, ordinal encoding the smallest parity gap and the baseline the smallest equalized-odds gap. The mean parity gap differed by 0.0657 between the best and worst pipeline, so these choices shift fairness but do not remove the gap. The undersampling and ordinal results should be read carefully, because each changes more than one thing at once.

The contribution is a repeatable, paired comparison of preprocessing choices with uncertainty estimates, extending earlier work on data preparation and fairness to a new classifier and to feature scaling. The study is limited to one dataset from 1994, one classifier and binary sex, and it did not cover binning, target encoding or skew transforms. Practitioners should therefore treat preprocessing as one lever among several, and combine it with explicit fairness measurement and, where needed, mitigation applied to the data, the model or its outputs.

7.2 Future Scope

1. Expand the transformation grid: Add equal-width, equal-frequency and domain-informed binning, target encoding, robust scaling, and log, Box–Cox and Yeo–Johnson transforms for capital-gain/loss. Use factorial experiments and ANOVA to quantify each factor’s effect on the parity gap.
2. Use newer census data: Test the five pipelines on ACSIncome across multiple years and states to assess temporal and geographic consistency.
3. Add classifiers: Repeat 30 splits using Random Forest and Gradient Boosting to compare preprocessing effects across model families.
4. Combine with mitigation: Pair each pipeline with reweighing or equalized-odds post-processing to test their combined effects on fairness.
5. Evaluate intersectional fairness: Analyze sex × race groups and report worst-group selection rates and Disparate Impact Ratio (DIR).
6. Isolate undersampling effects: Use a 2×2 design balancing labels, sex composition, both or neither, plus threshold retuning, to identify the cause of F5’s F1 improvement.

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APPENDIX A: IMPORTANT CODE

Project Title: Preprocessing Choices and Fairness: A Controlled, Repeated-Split Study on UCI Adult Income
Source Code File: fairness_preprocessing_F1_F5.ipynb
Description: The source code implements dataset acquisition, data cleaning, preprocessing pipelines (F1–F5), repeated train-test splitting, Logistic Regression training, performance and fairness evaluation, bootstrap confidence intervals, and visualization of the experimental results.
Source Code : https://github.com/aqdas07/fairness-preprocessing-project

Appendix B: Dataset Samples

Dataset link: https://archive.ics.uci.edu/dataset/2/adult (UCI Machine Learning Repository, reference [6]). It is downloaded with fetch_ucirepo(id=2), so there is no Web API key.
Raw dataset (Before cleaning):-

	row	age	workclass	education	occupation	sex	hours-per-week	native-country	income
0	0	39	State-gov	Bachelors	Adm-clerical	Male	40	United-States	<=50K
1	1	50	Self-emp-not-inc	Bachelors	Exec-managerial	Male	13	United-States	<=50K
2	2	38	Private	HS-grad	Handlers-cleaners	Male	40	United-States	<=50K
3	3	53	Private	11th	Handlers-cleaners	Male	40	United-States	<=50K
4	4	28	Private	Bachelors	Prof-specialty	Female	40	Cuba	<=50K
5	5	37	Private	Masters	Exec-managerial	Female	40	United-States	<=50K
6	6	49	Private	9th	Other-service	Female	16	Jamaica	<=50K
7	7	52	Self-emp-not-inc	HS-grad	Exec-managerial	Male	45	United-States	>50K
8	8	31	Private	Masters	Prof-specialty	Female	50	United-States	>50K
9	9	42	Private	Bachelors	Exec-managerial	Male	40	United-States	>50K
14	14	40	Private	Assoc-voc	Craft-repair	Male	40	?	>50K
27	27	54	?	Some-college	?	Male	60	South	>50K
38	38	31	Private	Some-college	Sales	Male	38	?	>50K
51	51	18	Private	HS-grad	Other-service	Female	30	?	<=50K
61	61	32	?	7th-8th	?	Male	40	?	<=50K

Cleaned dataset:-

	row	age	workclass	education	occupation	sex	hours-per-week	native-country	income
0	0	39	State-gov	Bachelors	Adm-clerical	Male	40	United-States	0
1	1	50	Self-emp-not-inc	Bachelors	Exec-managerial	Male	13	United-States	0
2	2	38	Private	HS-grad	Handlers-cleaners	Male	40	United-States	0
3	3	53	Private	11th	Handlers-cleaners	Male	40	United-States	0
4	4	28	Private	Bachelors	Prof-specialty	Female	40	Cuba	0
5	5	37	Private	Masters	Exec-managerial	Female	40	United-States	0
6	6	49	Private	9th	Other-service	Female	16	Jamaica	0
7	7	52	Self-emp-not-inc	HS-grad	Exec-managerial	Male	45	United-States	1
8	8	31	Private	Masters	Prof-specialty	Female	50	United-States	1
9	9	42	Private	Bachelors	Exec-managerial	Male	40	United-States	1
14	14	40	Private	Assoc-voc	Craft-repair	Male	40	Unknown	1
27	27	54	Unknown	Some-college	Unknown	Male	60	South	1
38	38	31	Private	Some-college	Sales	Male	38	Unknown	1
51	51	18	Private	HS-grad	Other-service	Female	30	Unknown	0
61	61	32	Unknown	7th-8th	Unknown	Male	40	Unknown	0

EVALUATION RUBRIC

No.	Evaluation Criterion	Max Marks	Obtained	Grade	Remarks
1	Problem Understanding and Literature Review	10			
2	Data Acquisition (sources, APIs, scraping)	10			
3	Data Cleaning and Munging Quality	10			
4	Missing Data Handling and Imputation	10			
5	Outlier Detection and Treatment	05			
6	Data Transformation and Feature Engineering	10			
7	Visualization Quality and Design	15			
8	Dashboard Development (Power BI / Tools)	10			
9	Results, Analysis and Research Contribution	10			
10	Report Quality and Presentation	05			
11	Publication Readiness (Appendix D Draft)	05			
	TOTAL	**100**			

Project Guide **Internal Examiner** **External Examiner**